SOS Bit Problem

Time limit: 1.00 s **Memory limit:** 512 MB

Given a list of n integers, your task is to calculate for each element x:

1. the number of elements y such that $x \mid y = x$
2. the number of elements y such that $x \& y = x$
3. the number of elements y such that $x \& y \neq 0$

Input

The first line has an integer n: the size of the list.

The next line has n integers $x_1, x_2, \dots, x_n$: the elements of the list.

Output

Print n lines: for each element the required values.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le x_i \le 10^6$

Example

Input	Output
5 3 7 2 9 2	3 2 5 4 1 5 2 4 4 1 1 3 2 4 4